

Question:

Carlos is planting a lemon tree. The tree will cost \$90 to plant. Each year it will grow 7 lemons, which he can sell for \$1.5 each. It costs \$3 a year to water and feed the tree. How many years will it take before he starts earning money on the lemon tree?

Reasoning + Answer:

<tool_call> Reasoning process here :

Step 1 : Understand the cost and revenue structure .

- Initial cost to plant the tree : \$ 90 (one-time cost).
- Each year , the tree produces 7 lemons .
- Each lemon sells for \$ 1 . 5 , so annual revenue = $7 \times 1.5 = \$ 10.50$.
- Annual cost to water and feed = \$ 3 .
- So , annual net profit = Revenue - Cost = $10.50 - 3 = \$ 7.50$.

Step 2 : Determine when Carlos starts earning money .

- The initial \$ 90 cost must be recovered before he starts earning a profit .
- Since he earns \$ 7 . 50 per year after the initial cost , we need to find how many years it takes to recover the \$ 90 cost .

Step 3 : Calculate the number of years to break even .

- Break-even point : Total earnings = Initial cost
- So : $7.50 \times n = 90$
- Solving for n : $n = 90 / 7.50 = 12$

Step 4 : Interpret the result .

- After 12 years , the total net profit will be \$ 90 , which covers the initial cost .
- Therefore , starting from year 13 , he will be earning money (since the cost is fully recovered and he starts making profit).
- But the question is : " How many years will it take before he starts earning money on the lemon tree ?"

Note : " Starts earning money " means after the initial cost is covered , so the first year where he makes a profit *after* paying the initial cost .

So , the answer is the number of years until the net profit covers the initial cost â that is , after 12 years , the cost is fully recovered , and from year 13 onward , he starts earning money .

But the question is : " How many years will it take before he starts earning money ?" This is asking when the *profit* becomes positive â i.e. , when the cumulative net profit covers the initial cost .

So , after 12 years , he has spent \$ 90 , and earned $7.50 \times 12 = \$ 90$ â so he breaks even .

Thus , after 12 years , he has not yet started *earning* money â he's just breaking even .

He starts *earning* money after the initial cost is recovered , so in year 13 .

But the question is : " How many years will it take before he starts earning money ?"

This is asking for the number of years *after* planting until he starts making a profit â meaning when the cumulative net profit exceeds the initial cost .

So , after 12 years , the net profit is exactly \$ 90 â he breaks even .

So , he starts earning money **after** 12 years â i.e. , in year 13 .

But the question is : " How many years will it take before he starts earning money ?"

So the answer is : **13 years**?

Wait â let's reconsider .

Actually , no â he starts earning money *after* the initial cost is covered . So after 12 years , he has paid \$ 90 , and earned \$ 90 in net profit â so he has broken even .

But " earning money " means making a profit â so when does he start making a *profit*?

After year 12 , he has not made a profit â he has just broken even .

So he starts making a profit in year 13 .

But the question is : " How many years will it take before he starts earning money ?"

So , after how many years does he start earning money ?

Answer : After 13 years? But he only earns \$ 7 . 50 per year .

But let's rephrase : after how many years does he start making a *profit*?

At the end of year 12 , he has broken even